

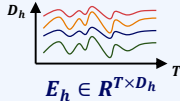
Input Interface

Multi-modal Inputs

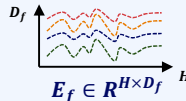
Historical Target Status



Historical Exogenous (ExG_h)



Future Exogenous (ExG_f)



Graph Metadata (M_G)

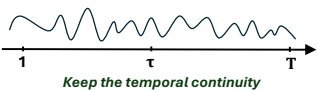


Adjacency / Coordinates / Attributes

AGSP Module

Adaptive Gaussian-soft-windowed Patching

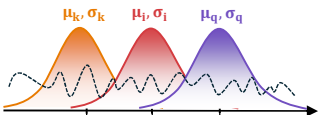
Continuous Time Axis



Soft Overlapping Gaussian Windows

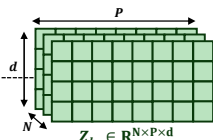
$$W_i(\tau) = \exp\left(-\frac{(\tau - \mu_i)^2}{2\sigma_i^2 + \varepsilon}\right)$$

$$\tilde{W}_i(\tau) = \frac{W_i(\tau)}{\sum_{j=1}^p W_j(\tau) + \varepsilon}$$



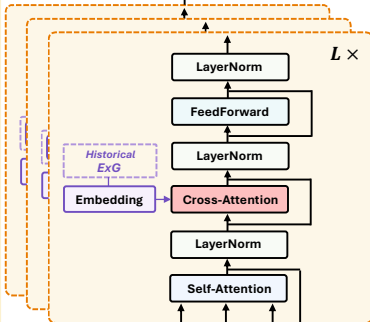
$$z_{b,n,i} = \varphi\left(\sum_{\tau=1}^T \tilde{W}_i(\tau) \bar{x}_{b,\tau,n}\right) + e_n + p_i$$

Weighted Aggregation



Temporal Transformer

Multi-head Self-Attention with Global Token



Add & Norm



Add Global Token



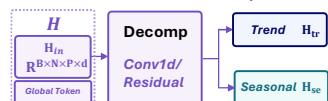
Input: Node-patch Tokens from AGSP

R-STMF Module

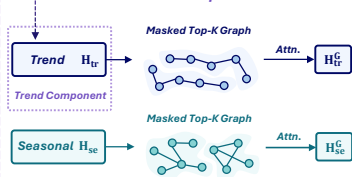
Relational Spatio-Temporal Manifold Fusion

$$H \in \mathbb{R}^{B \times N \times (P+1) \times d}$$

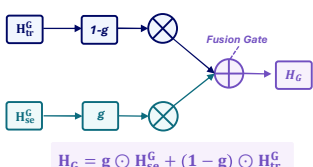
Latent Patch Manifold Decomposition



Distance-Masked Graph Interaction



Gated Fusion



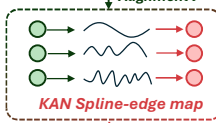
$$H_G \in \mathbb{R}^{B \times N \times (P+1) \times d}$$

K-AMC Module

KAN-based Exogenous Modulation

$$\text{Future ExG } E_f \in \mathbb{R}^{H \times D_f}$$

Alignment Γ



γ, β

FILM Modulation

$$H_{mod} = \gamma \odot H_G + \beta$$

Convex Blend

$$H_F = (1 - \alpha) H_G + \alpha H_{mod}$$

$$H_F \in \mathbb{R}^{B \times N \times (P+1) \times d}$$

PI-DCH Module

Physics-Informed Dynamic Capping

Linear Decoder

Barrier Gate

Soft Capping

Cons. Ω, C_{dyn}

Strict Projection

Network Forecast $\tilde{Y} \in \mathbb{R}^{B \times H \times N \times D_y}$